1D diamond nanobeam optomechanical crystal for the SnV center

Study summary — Abby — generated 2026-10-15

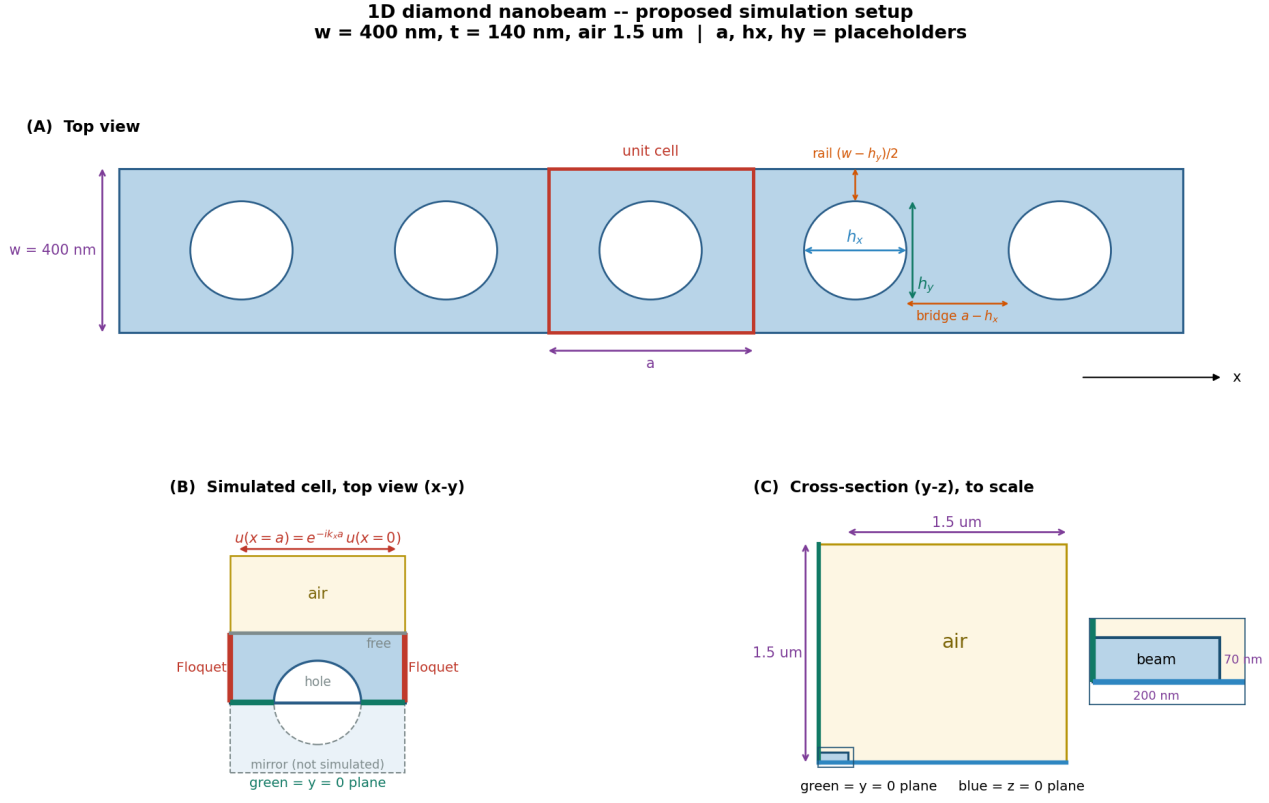
Goal. A diamond nanobeam cavity whose fundamental TE-like optical mode sits at the SnV zero-phonon line (619 nm, 484 THz, n = 2.41) and whose co-localized mechanical breathing mode sits at or below 10 GHz, following the design approach of Oh et al., Optica 13, 485 (2026): symmetry-protected optical and mechanical *quasi*-band gaps in an elliptical-hole mirror cell, with a smooth taper to a central defect cell (the Chan, Caltech PhD thesis 2012 parameterization). A complete mechanical gap (for Q_m) comes later from a 2D shield.

Tools. COMSOL 6.4 driven from Python (mph) on the lab Windows machine; isotropic diamond (E = 1050 GPa, ν = 0.07, ρ = 3500 kg/m³); Lumerical FDTD for independent optical verification (later).

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1. Mirror-cell model

One period of the nanobeam (period a, width w, thickness t, elliptical hole hx × hy), modeled as a quarter cell with Floquet periodicity along x and symmetry planes at y = 0 and z = 0 (scripts/build_nanobeam_model.py). Mechanics on the beam only, in four vector-symmetry classes (Symmetry/Antisymmetry on each plane: Y±Z±); Y+Z+ is the breathing class. Optics on beam + 1.5 μm air: TE-like class = PEC on y = 0, PMC on z = 0; guided modes below the light line f = ck/2π.



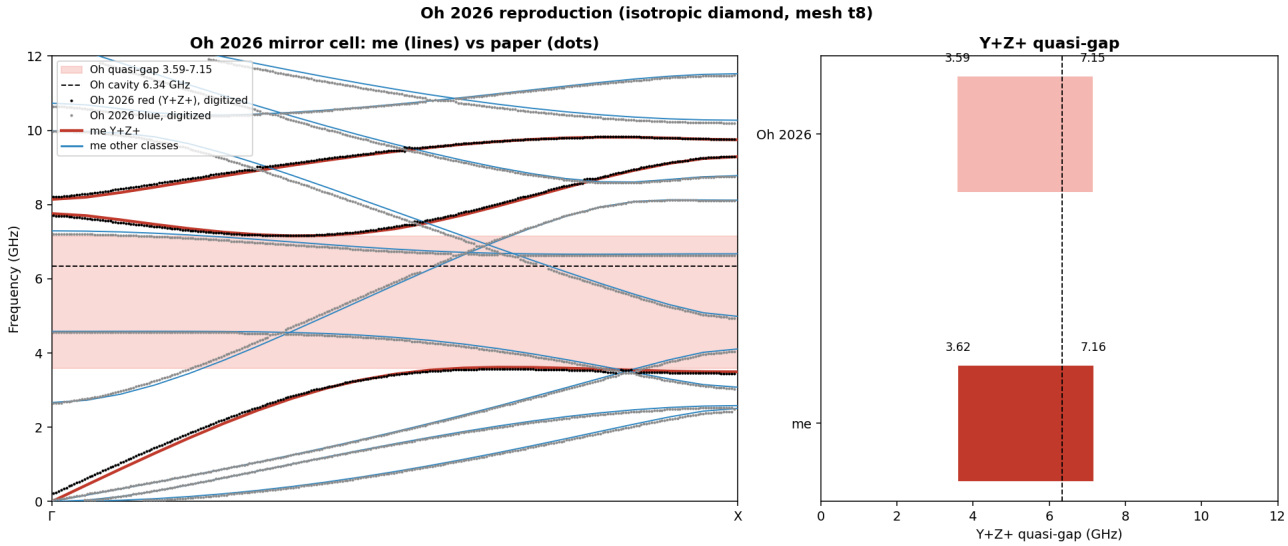
class	y = 0	z = 0	mode family
Y+ Z+	sym	sym	longitudinal / breathing
Y- Z+	antisym	sym	in-plane bending
Y+ Z-	sym	antisym	out-of-plane bending
Y- Z-	antisym	antisym	torsion

Mirror-cell model: geometry, domains and boundary conditions.

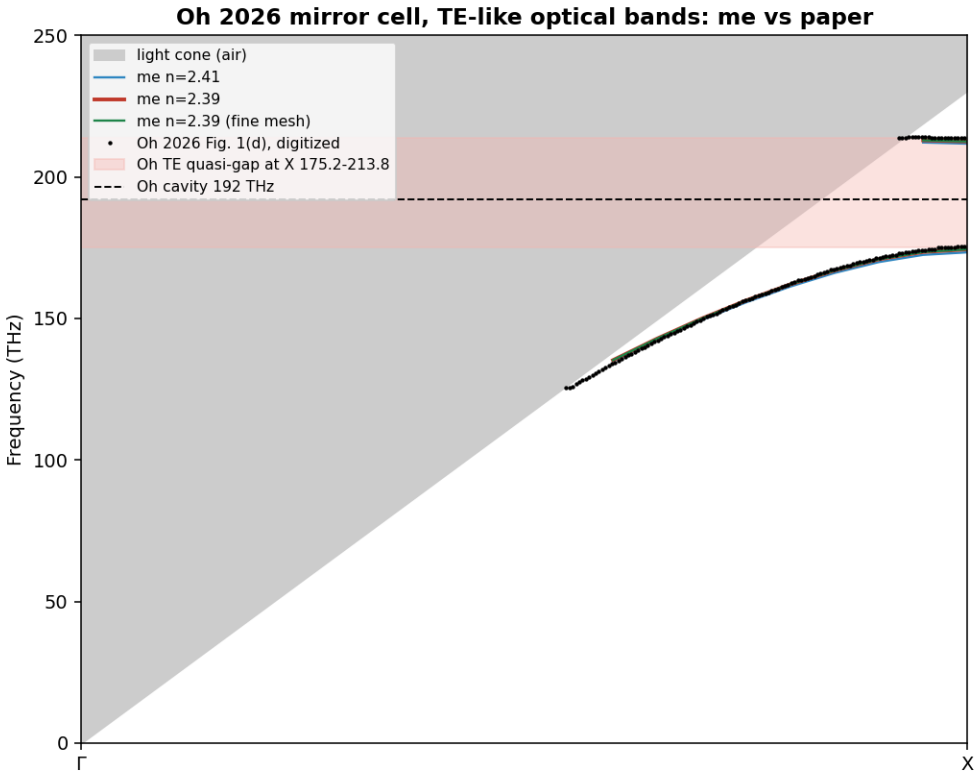
2. Validation: Oh 2026 mirror cell

Oh mirror cell (a, w, hx, hy, t) = (650, 800, 343, 617, 250) nm, compared with Oh Fig. 1(d, e) digitized point by point.

	me	Oh 2026
Y+Z+ mechanical quasi-gap (GHz)	3.62 - 7.16	3.59 - 7.15
TE band edges at X (THz), n = 2.39	174.4 / 212.5	175.2 / 213.8
mesh convergence (t/5 vs t/8, mech)	median 5e-4	
optical mesh convergence	6e-5	



Mechanical bands, me vs Oh Fig. 1(e).



TE optical bands, me vs Oh Fig. 1(d).

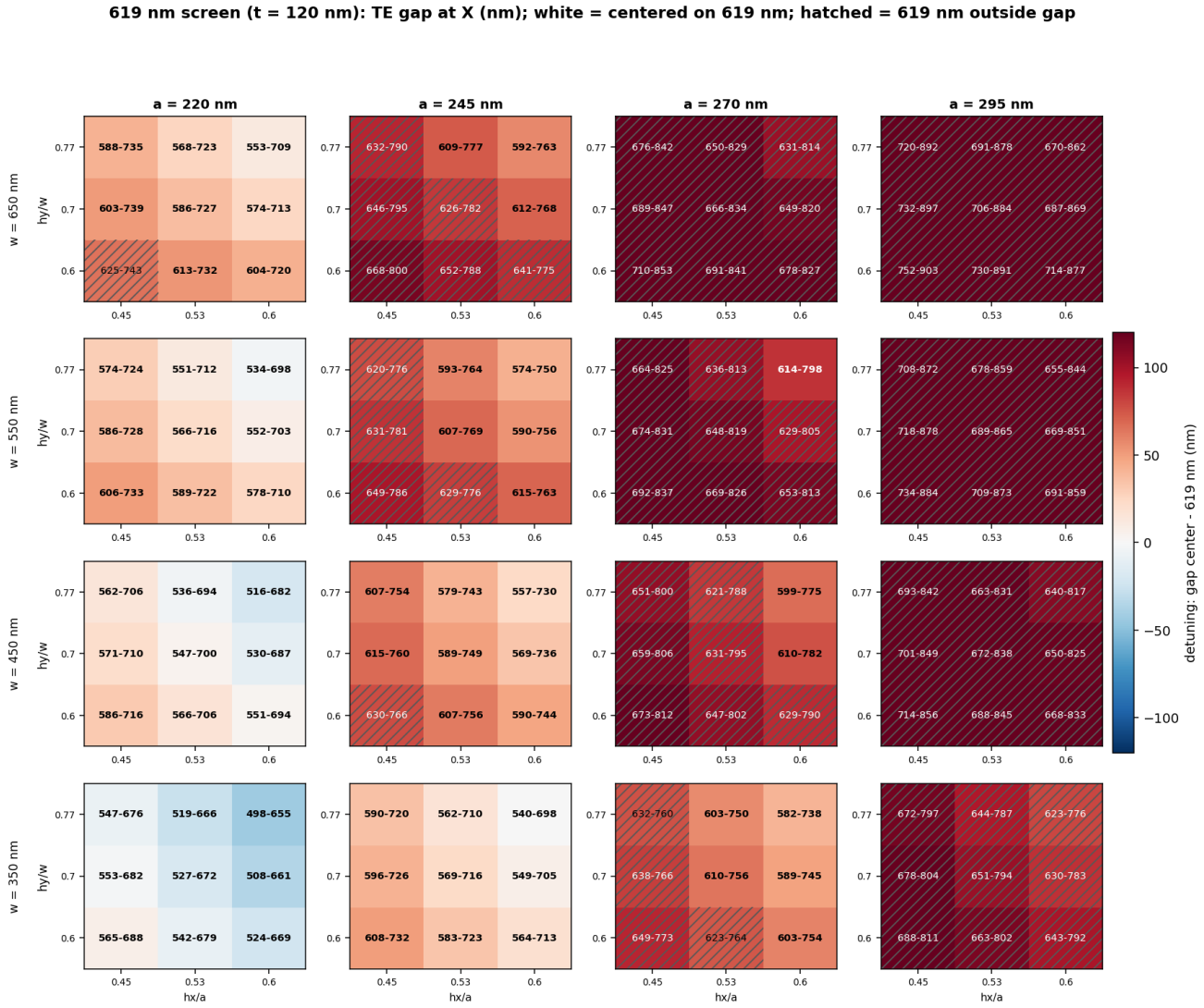
2b. Elastic model (isotropic vs cubic anisotropic)

Oh does not state his elastic model. Test: mirror-cell bands with cubic diamond (C11, C12, C44 = 1076, 125, 578 GPa), beam along [100] and along [110], vs every digitized point. Isotropic so far: median error 45 MHz (Y+Z+), 50 MHz (others).

Pending: anisotropic [100] / [110] band comparison (script runs/nanobeam_oh2026/bands_aniso.py; parked)

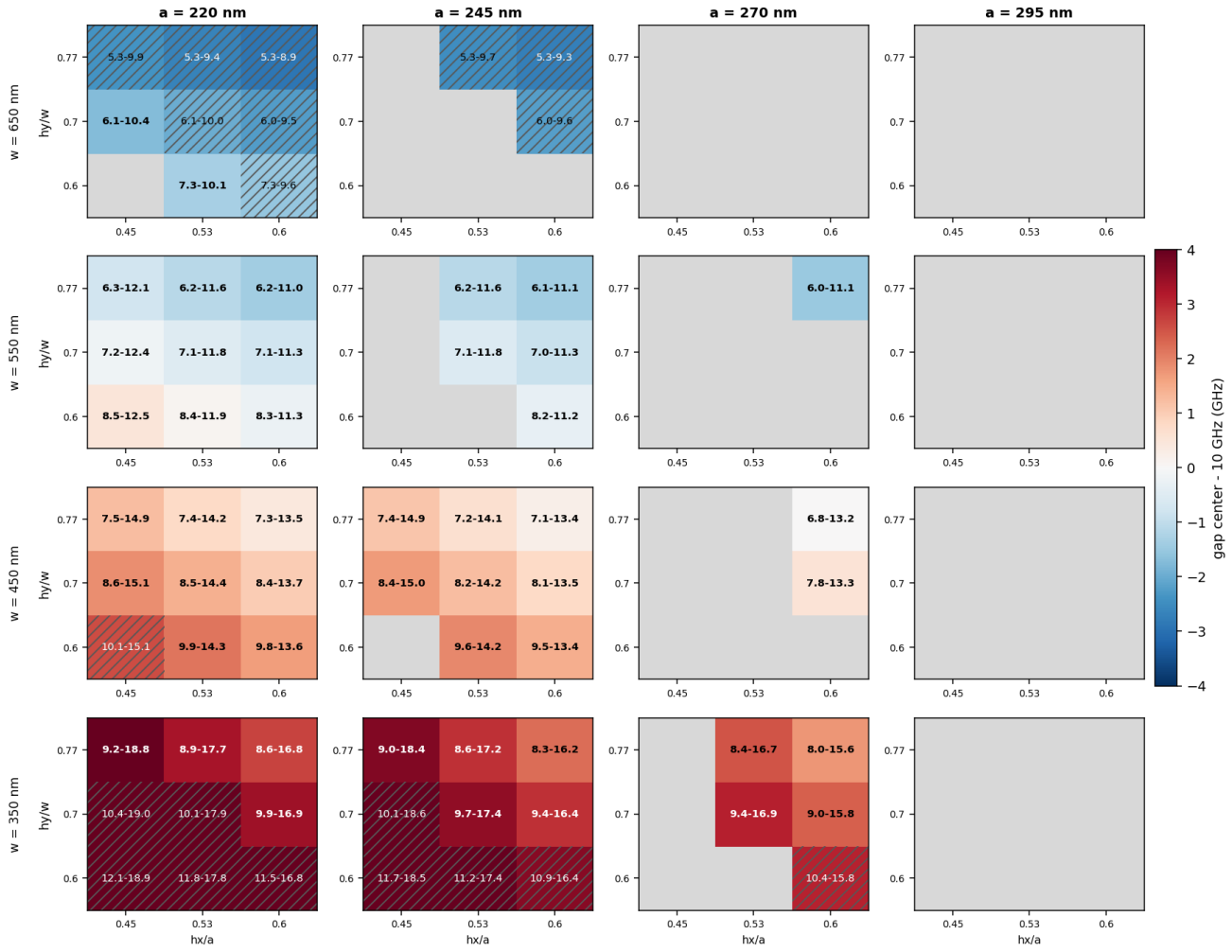
3. Mirror-cell design at 619 nm (t = 120 nm)

Stage 1 (optics at X). 144 cells: w 350-650, a 220-295, hx/a 0.45-0.60, hy/w 0.60-0.77. 68 contain 619 nm in the TE gap.

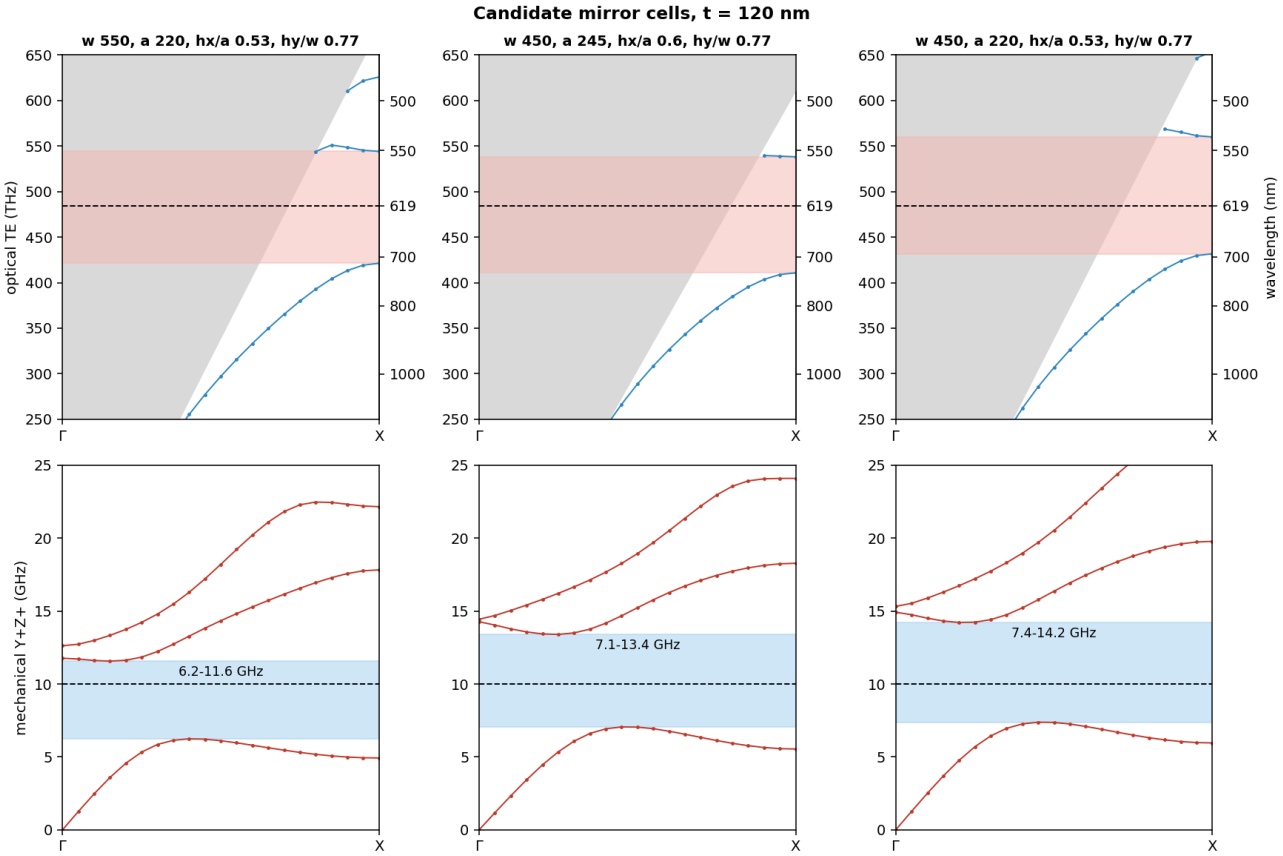


TE gap at X (nm); color = gap center - 619 nm; hatched = 619 nm outside the gap.

Stage 2 (breathing-class mechanics). Y+Z+ bands for the 68 cells; 48 also hold 10 GHz in the band 1-2 quasi-gap. The beam width sets the gap: w 650 ~5-10 GHz, w 550 ~6-12.5 GHz, w 450 ~7-15 GHz, w 350 ~9-19 GHz.



Y+Z+ quasi-gap (bands 1-2), GHz; color = gap center - 10 GHz; hatched = 10 GHz outside.



Optical (top) and mechanical Y+Z+ (bottom) bands of the three candidate cells.

Favored cell: **w 550, a 220, hx/a 0.53, hy/w 0.77** (breathing gap 6.2-11.6 GHz, optical 551-712 nm). Its breathing band at Γ (11.7 GHz) needs only ~15 % pull to reach 10 GHz.

4. Full-cavity method

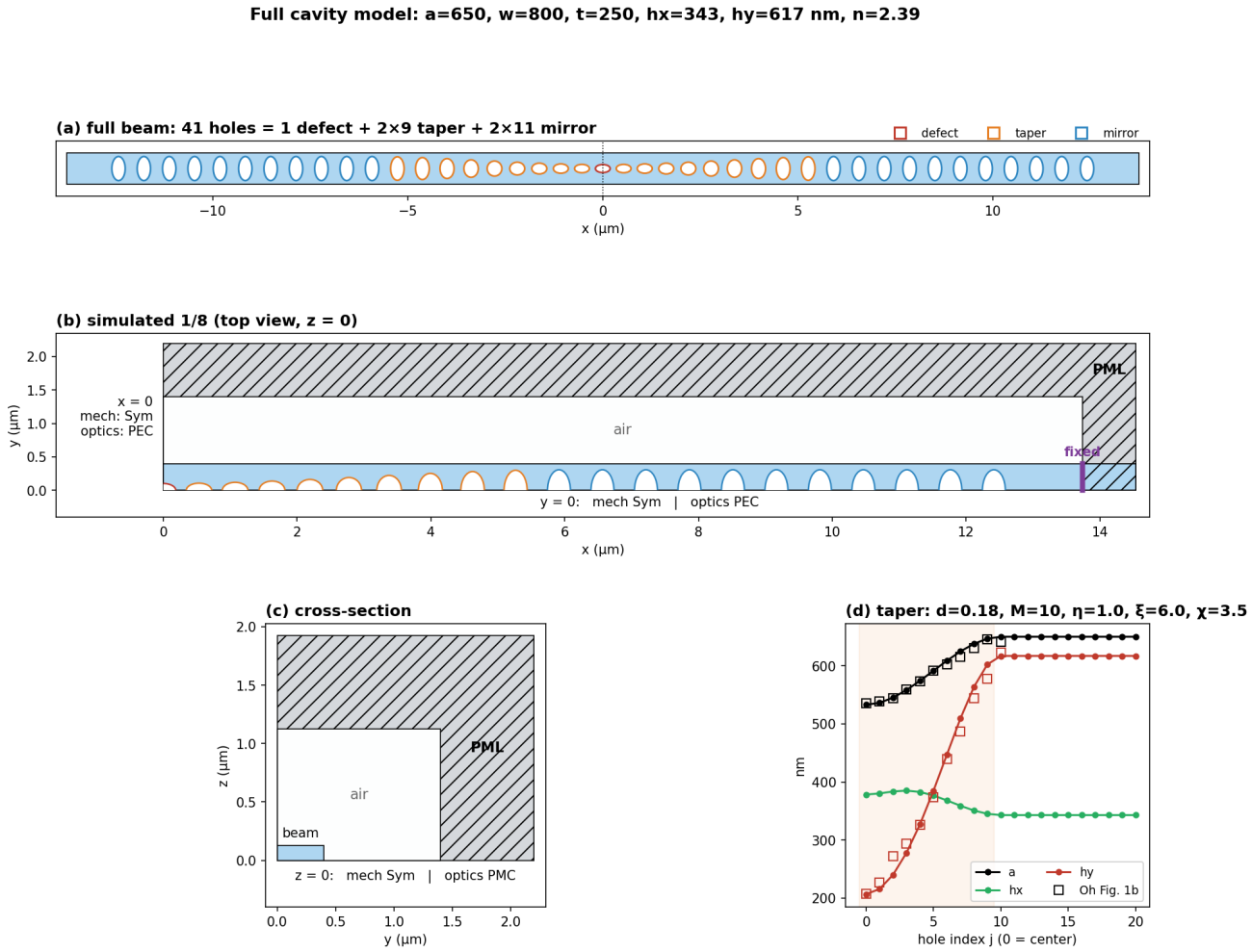
Cells j = 0 (defect, beam center) ... M (mirror), then n_{mirror} mirror cells (Chan thesis §3.4; Oh Supplement S3):

- period: $a_j/a_m = 1 - d\,(2y^3 - 3y^2 + 1)$, y = η-remap(j/M)
- hole area: $1 - s_j = \chi\,(1 - a_j/a_m)$; hole aspect (hy/hx): $r_j = (a_j/a_m)^{\xi}$
- optical cavity mode: X-point dielectric band pushed up into the gap by the smaller defect period
- mechanical cavity mode: Y+Z+ breathing band at Γ pulled down by flatter center holes
- diamond photoelastic constants (p11, p12, p44) = (-0.25, 0.043, -0.172) (Burek 2016, citing Lang 2009)

Oh taper measured from his Fig. 1(b) (45 nm/pixel): 41 holes = center + ~10 taper + ~10 mirror cells per side; Chan cubic fit d = 0.177, M = 11.65, η = 1.05 (rms 0.005); center hy ≈ 208 nm, hx ≈ 400 ± 45 nm → ξ ≈ 6, χ ≈ 3.5. Optical E_y is odd about the center hole (x = 0 node).

5. Full-cavity model

One-eighth of the beam (scripts/build_cavity_model.py): symmetry planes x = 0 (mech Symmetry / optics PEC), y = 0 (Symmetry / PEC), z = 0 (Symmetry / PMC); air + PML on +x, +y, +z; the beam runs into the +x PML (light through the mirrors is absorbed); mechanics fixed at the beam end; one shared mesh for both physics. Optical Q from the complex eigenfrequency; cavity mode = most localized mode with Q > 10^3; breathing mode = localized, in-phase sidewall motion and large sidewall amplitude (section 9).

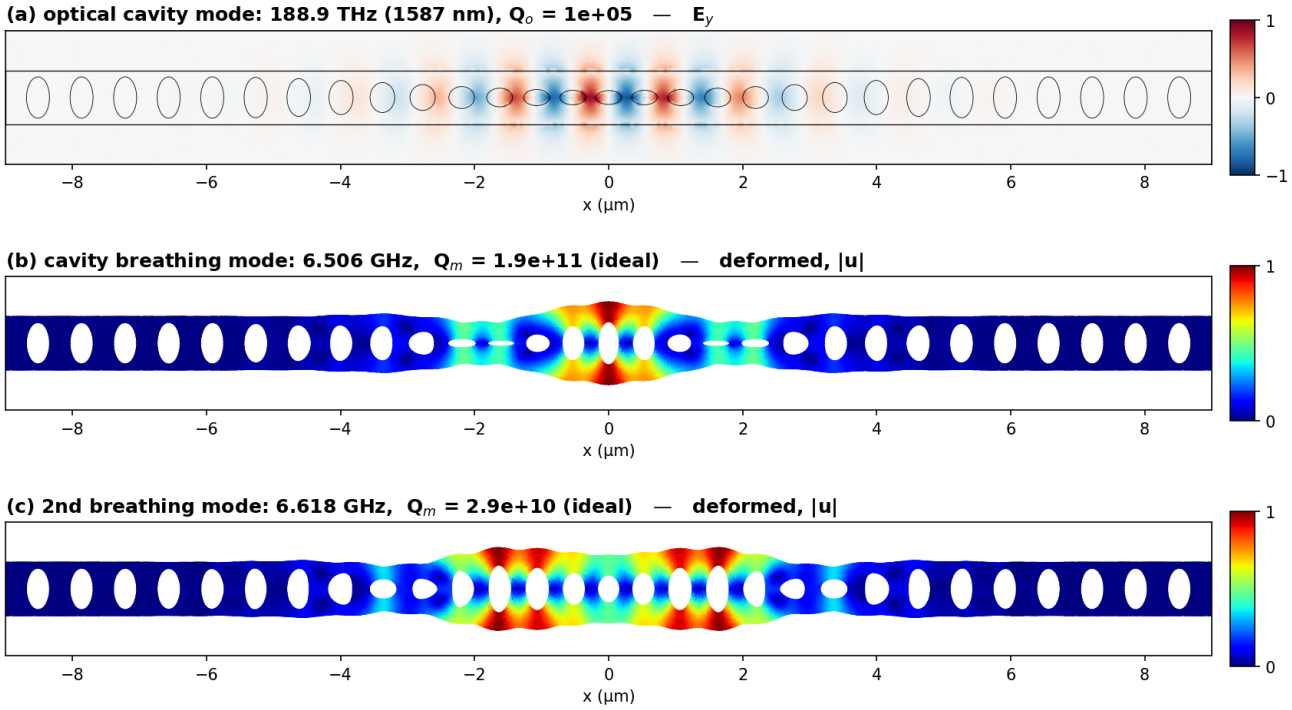


Full-cavity model (Oh 2026 cavity shown).

6. Validation: Oh 2026 cavity

	me	Oh sim	Oh measured
optical mode (THz)	188.9	192	190.1
breathing mode (GHz)	6.51	6.34	6.23
optical Q	1.0e5	-	3.8e4
mechanical Q	1.9e11 (ideal geometry)	not simulated	1.9e6

Oh 2026 cavity reproduction, me (|x| < 9 µm of the 25 µm beam)

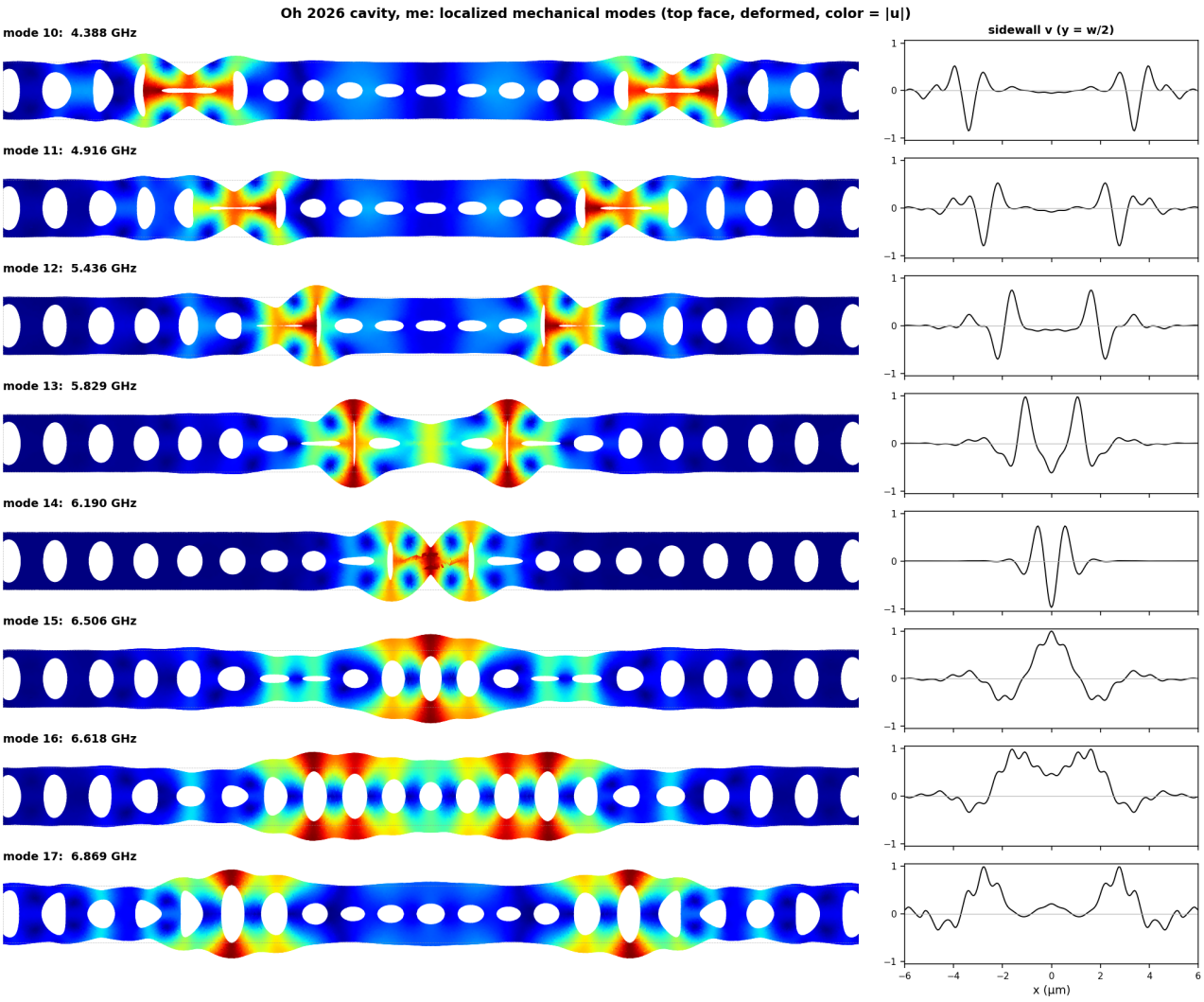


(d) vs Oh 2026

	me	Oh sim	Oh meas
optical (THz)	188.9	192	190.1
mech. (GHz)	6.51	6.34	6.23
Q_o	1.0e+05	—	3.8e4
Q_m	2e+11*	not sim.	1.9e6

* ideal geometry, no disorder, no shield

Oh cavity: optical E_y (a), first and second breathing modes on the deformed beam (b, c), comparison (d).



All localized mechanical modes (deformed top face) and their sidewall motion; modes 15-16 breathe.

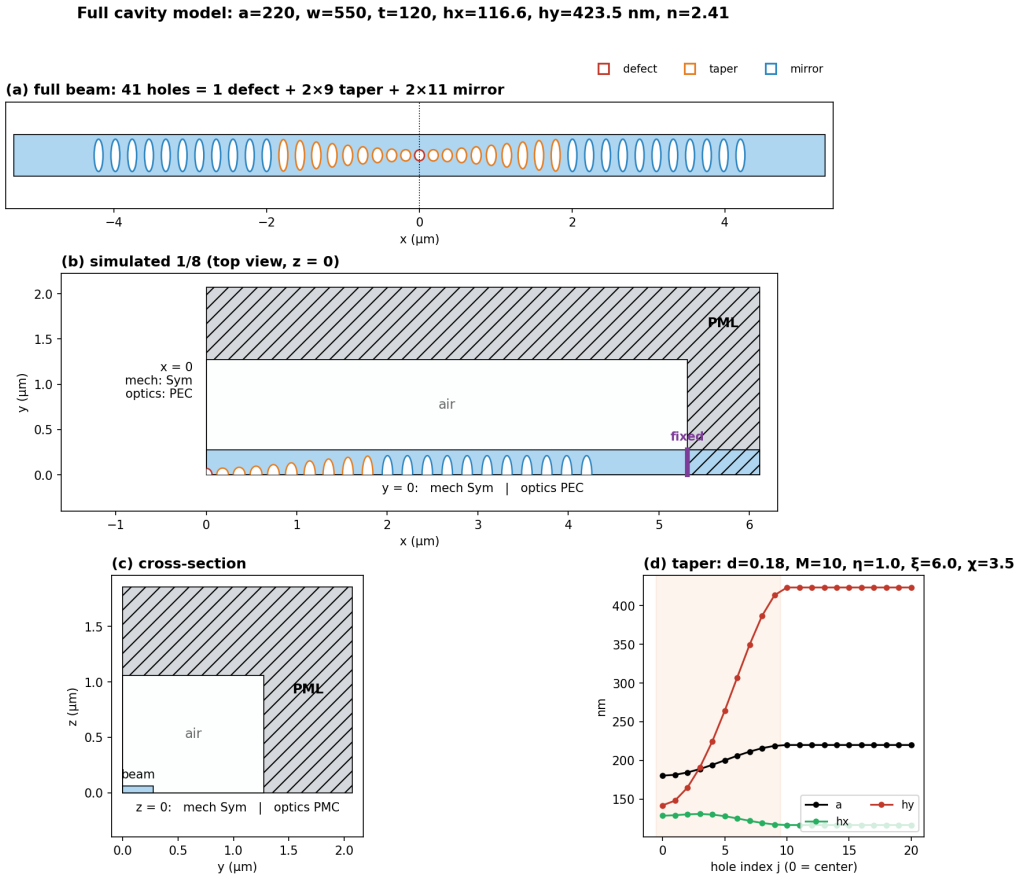
6b. Mechanical Q benchmark

Ideal symmetric geometry gives Q_m ~ 10^{10}-10^{11} (quasi-gap protection). Realistic Q_m needs symmetry breaking (fabrication disorder) and Oh's cross-shaped acoustic shield (SEM pitch ~0.8 μm).

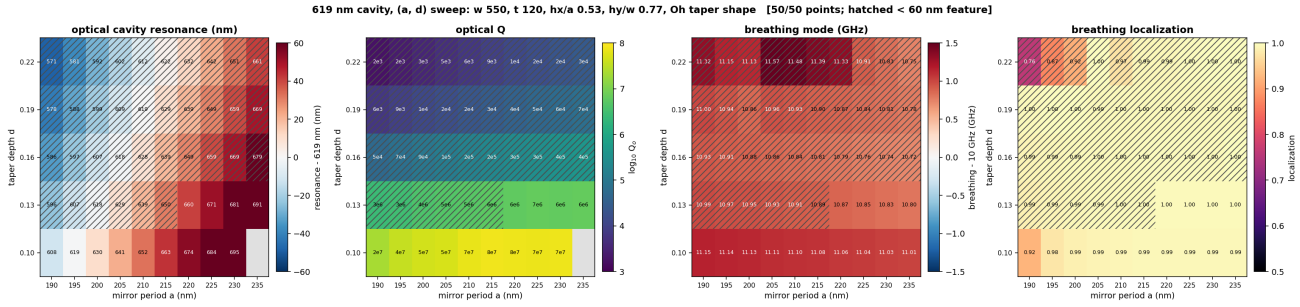
Pending: Q_m vs hole disorder (σ = 1, 2, 4, 8 nm), without and with the cross shield

7. 619 nm cavity: (a, d) sweep

Favored mirror cell (w 550, t 120, hx/a 0.53, hy/w 0.77) with Oh's taper shape (M 10, η 1, ξ 6, χ 3.5, 10 mirror cells); mirror period a 190-235 nm × taper depth d 0.10-0.22. Air/PML box reduced to 600/500 nm after a check (resonance change 2.4e-6, Q change 0.4 %).



619 nm cavity template (a 220, d 0.18).



Optical resonance (nm), optical Q, breathing frequency (GHz), breathing localization. Hatched: a feature below 60 nm.

619 nm line (interpolated along a):

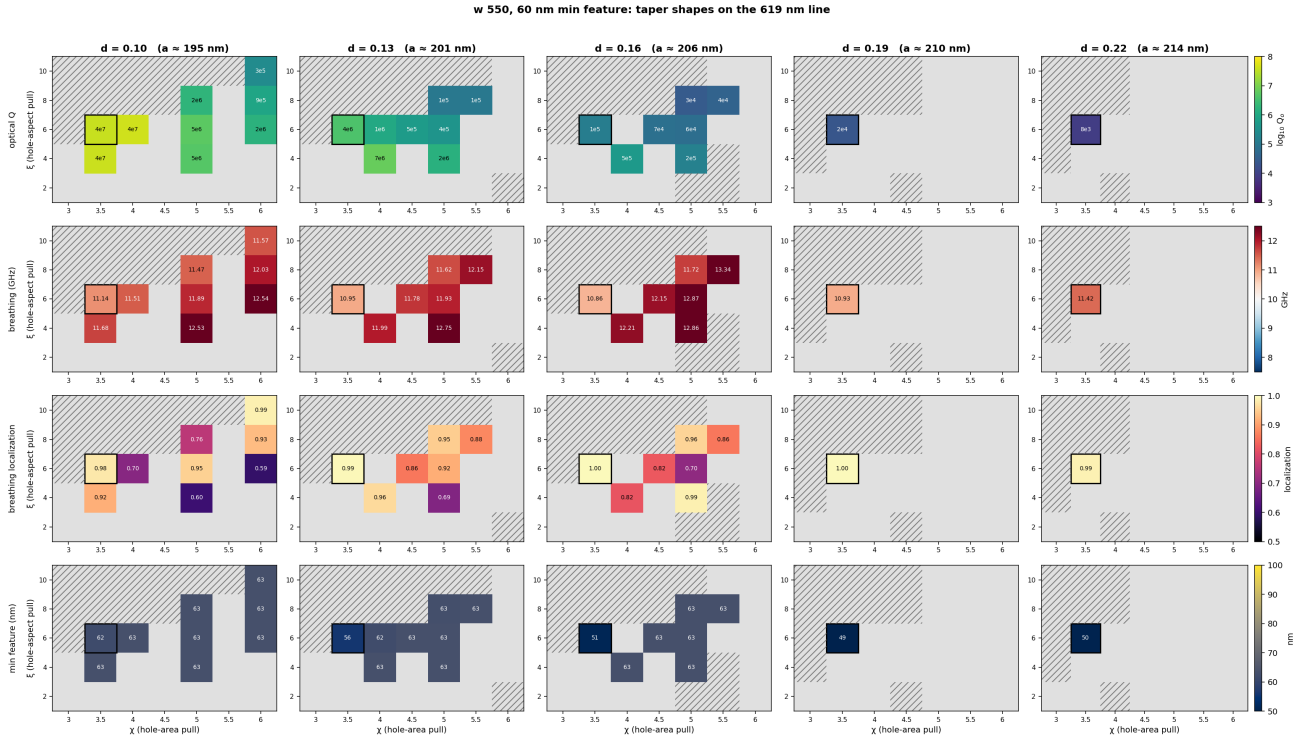
taper depth d	a for 619 nm	optical Q	breathing (GHz)	fabricable at 60 nm
0.10	195.2 nm	3.7e7	11.14	yes
0.13	200.7 nm	4.1e6	10.95	no (bridges 51-56 nm)
0.16	205.7 nm	~1.3e5	10.86	no
0.19	210.1 nm	~2e4	10.93	no

0.22	213.7 nm	~8e3	11.42	no

Findings: a and d set the resonance (~2 nm per nm of a; deeper d shortens it); shallower tapers give much higher Q; the breathing mode stays at 10.7-11.6 GHz everywhere and is well localized (0.98-1.00 except at the grid corners). Best point: d 0.10, a 195.2: 619.35 nm, Q 3.0e7 (confirmation solve), breathing 11.14 GHz, 62 nm minimum feature.

8. Taper-shape sweep (minimum feature 60 nm)

Center-hole aspect ξ and area χ at d = 0.10, 0.13 and 0.16 on the 619 nm line (2-3 periods per shape, interpolated to 619 nm); every point checked against the 60 nm minimum feature. Goal: breathing mode ≤ 10 GHz at the highest optical Q.



Taper shapes on the 619 nm line, one column per d: optical Q, breathing frequency, breathing localization, minimum feature. x = χ, y = ξ; outlined = Oh shape from the (a, d) step; hatched = below 60 nm; grey = not run.

Result. No fabricable taper shape brings the breathing mode below ~10.9 GHz at w = 550; stronger pulls mostly cost optical Q or mechanical localization. Best w 550 device: the (a, d) design d 0.10, a 195.2 (Q 3.0e7, 11.14 GHz). The taper is not enough: widen the beam (breathing frequency ~ 1/w; section 10).

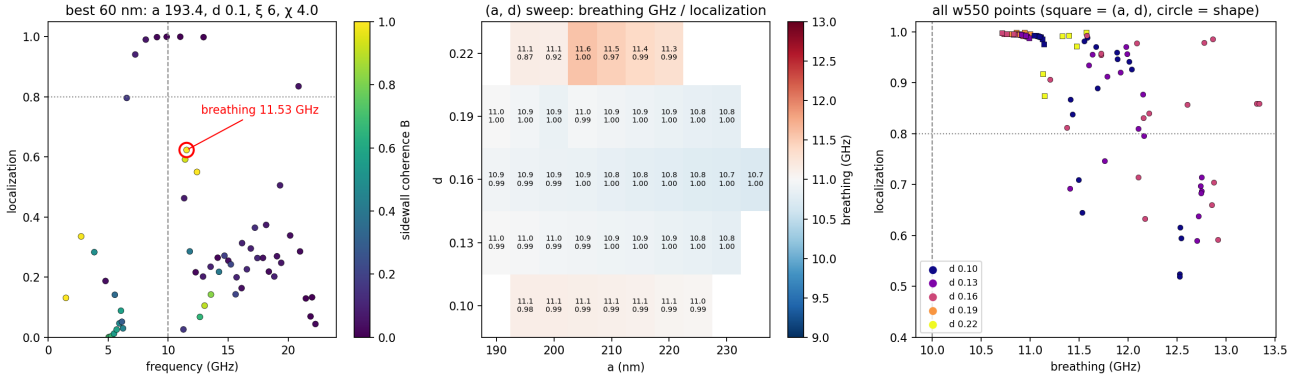
9. Breathing-mode identification

Three checks, refined as the sweeps exposed failure modes (2026-10-15):

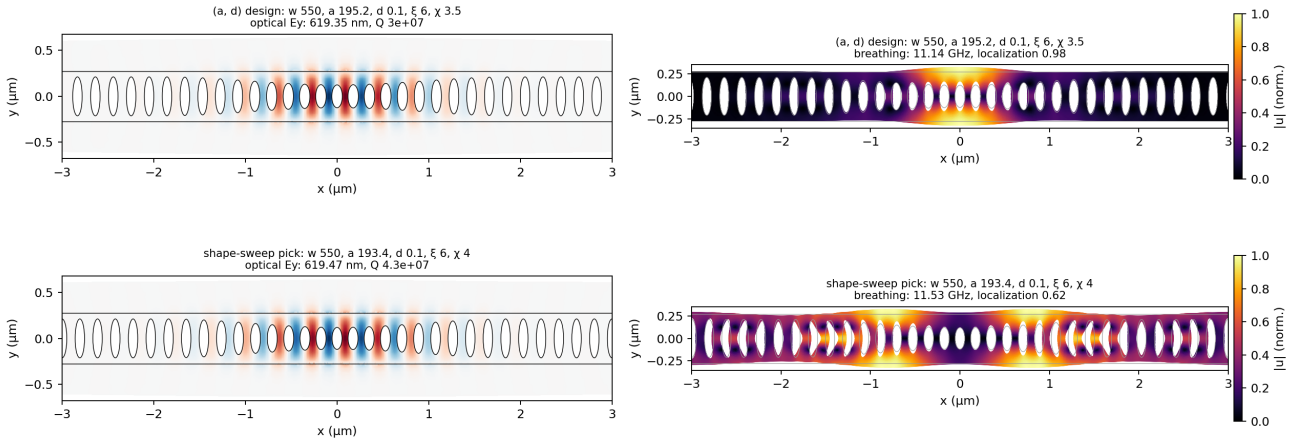
• **localization** (fraction of |u|² within 1.5 μm of the center) ≥ 0.8;

• **sidewall coherence** B > 0.5 (the two sidewalls move in phase). The earlier "largest optomechanical overlap among localized modes" rule picked weak non-breathing modes whenever the breathing mode fell outside the eigenvalue window (a false 8.2 GHz result); the window is now 60 modes around 11 GHz;

• **sidewall amplitude ratio** S = max sidewall |v| / max |u| ≥ 0.7. B is amplitude-blind: at w 650, d 0.13 a hole-centered hybrid (S = 0.35) passed it. Clean breathing modes have S = 1.0. Stored for every solve from now on; older points get it from their field maps and from the confirmation solves (section 11).



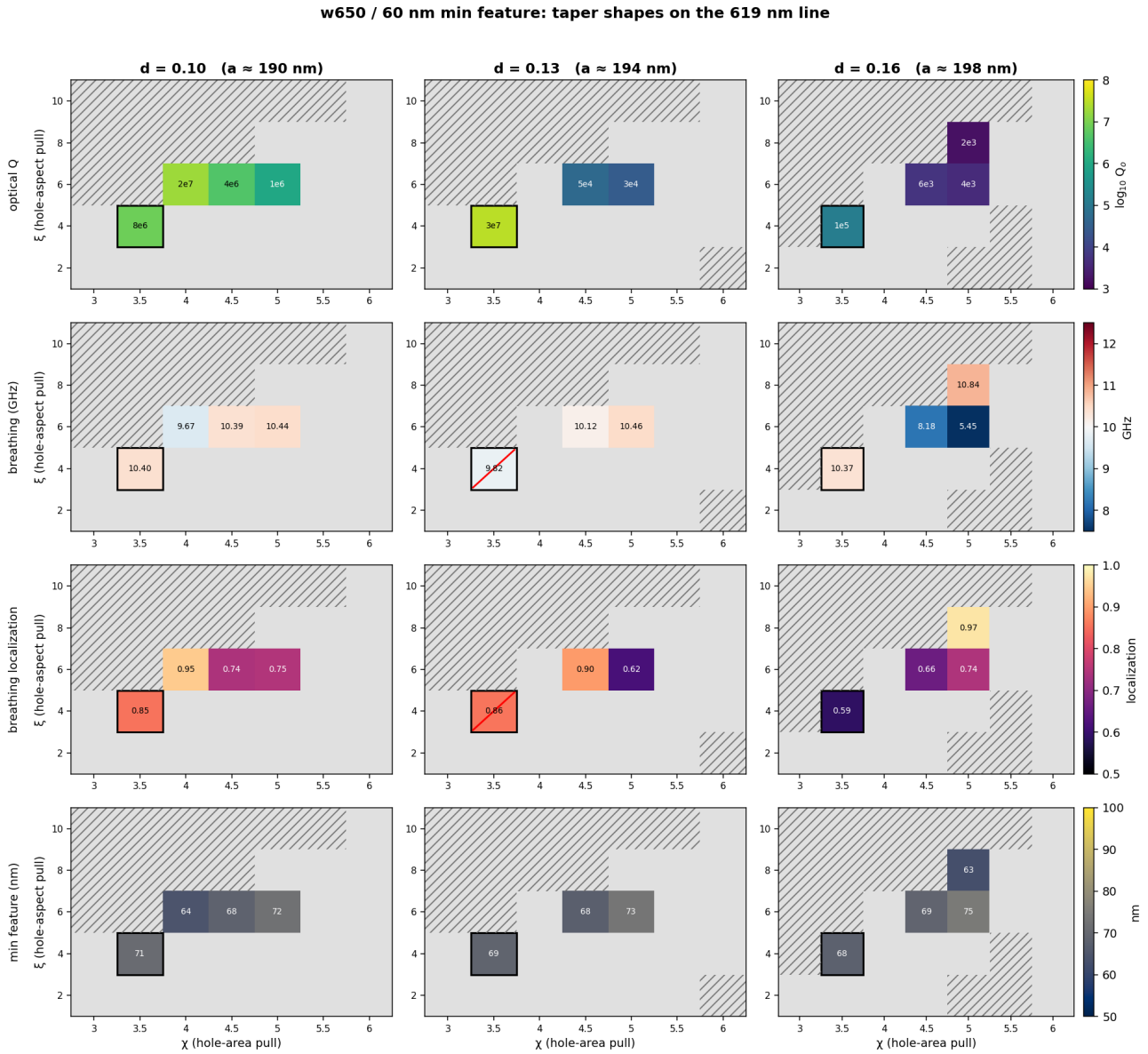
w 550: every mechanical mode of the best device (frequency vs localization, color = B); (a, d) map of the breathing frequency / localization; all points.



w 550: optical E_y and breathing mode on the deformed beam. Top: clean (a, d) design. Bottom: a shape variant whose "breathing" mode is a double-lobed mode leaking into the mirrors (localization 0.62).

10. Beam-width study (60 and 80 nm minimum feature)

Same mirror-cell ratios at w = 650 and 600 (650 first): 60 nm cell hx/a 0.53, hy/w 0.77; 80 nm cell hx/a 0.45, hy/w 0.70 (the earlier w 550 80 nm cell A, hx/a 0.53, had almost no fabricable tapers and a weakly localized breathing mode; dropped). Per d (0.10, 0.13, 0.16) the period is stepped until 619.4 nm is bracketed (base shape ξ 4, χ 3.5), then up to 8 taper shapes on the 619 nm line (runs/cavity_619_width/run.py).

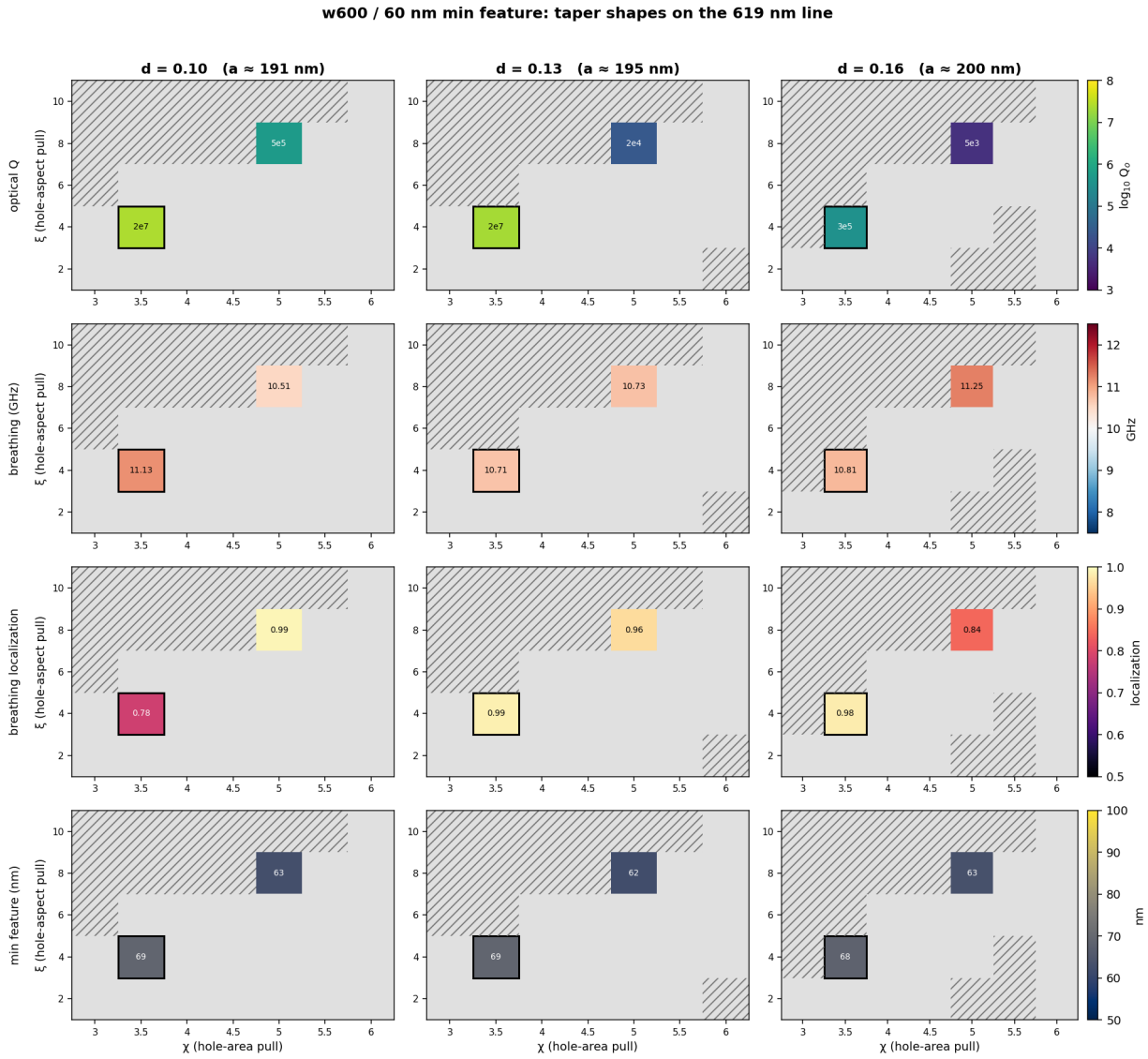


w 650, 60 nm: taper-shape maps (as section 8). Red slash = field map shows a hybrid, not a breathing mode.

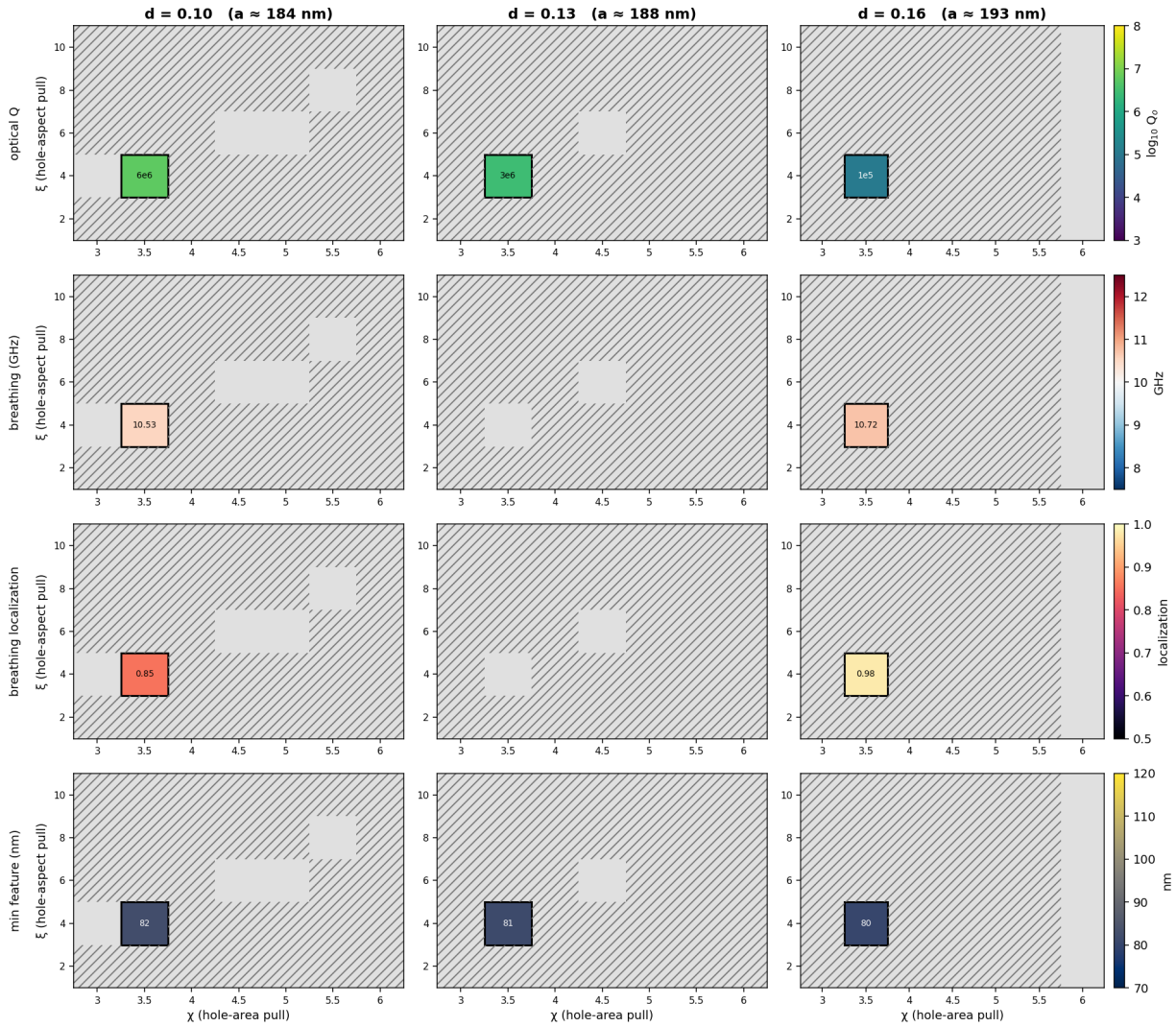
w 650, 60 nm. The breathing mode drops by ~1.3 GHz to 9.7-10.4 GHz with no loss of optical Q (3e7 at d 0.13, 1.8e7 at d 0.10 / ξ 6 / χ 4). Best: **d 0.10, ξ 6, χ 4.0, a 190.3: 619.50 nm, Q 1.8e7, breathing 9.67 GHz (localization 0.95, S 1.0), 64 nm minimum feature.** The d 0.13 base design (Q 3.0e7, 9.82 GHz) is a hybrid (S 0.35).

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w 650, 80 nm. 619 nm needs a 184-193 nm; there no taper shape beyond the base keeps every bridge ≥ 80 nm. Best: base shape d 0.10, a 183.8: Q 5.7e6, breathing 10.53 GHz (localization 0.85): 0.5 GHz above target. Reaching ≤ 10 GHz at 80 nm likely needs a wider beam (w ~700) or a cell with smaller holes.



w 600, 60 nm: taper-shape maps.



w 650, 80 nm: taper-shape maps.

w600 / 80 nm
(pending)

w 600, 80 nm: taper-shape maps.

11. Selection and final devices (one per minimum feature)

Rule (scripts/cavity_designs.py): fabricable; breathing localization ≥ 0.8; not a field-checked hybrid; breathing ≤ 10 GHz first, highest optical Q; if none meets 10 GHz, lowest breathing frequency. The top 3 per pipeline are re-solved with the sidewall-amplitude check before the pick (runs/cavity_619_final/deliverables.py, after the width study).

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Final device, 60 nm minimum feature

Pending: confirmation solve, mirror-cell optical bands (both y-parities: single-mode check) and mechanical bands (4 classes), device field maps

Final device, 80 nm minimum feature

Pending: confirmation solve, mirror-cell optical bands (both y-parities: single-mode check) and mechanical bands (4 classes), device field maps

12. Lumerical FDTD verification

Independent optics check (full beam or 1/8 symmetry; Lumerical's Qanalysis method, as in the lab's nanobeam Q sweeps). All files in C:\Users\abby\lumerical. Built: lum_w650_mf60_best_check.fsp (current best 60 nm device: w 650, d 0.10, ξ 6, χ 4.0, a 190.3). Sweep replicas, point for point with COMSOL λ and Q alongside: lum_ad_mf60 (50), lum_shape_mf60 (52), lum_w650_mf60_ad (6), lum_w650_mf60_shape (17) (runs/cavity_619/lumerical/make_lsf.py). Run after COMSOL is idle, with use_symmetry = 1 and ~1.5 ps (Q ~ 10^7 cannot be resolved in practical FDTD time; it bounds Q from below).

Pending: .fsp files of the two final devices (build only; queued after the deliverables)

Pending: Lumerical vs COMSOL: resonance and Q heat maps for each sweep, field-map comparison

13. Open items

• 80 nm device ≤ 10 GHz: wider beam (w ~700) or a smaller-hole cell

• bridge-centered (even hole count) cavity: field maximum and breathing strain maximum both at the exact center (now both peak in the first bridge, ~30 nm apart, two mirror sites at x ≈ ±95 nm); test on the w 650 finalists

• optomechanical coupling (g_0) in the sweeps is mis-normalized (~10^8 too small); frequencies and Q unaffected; to debug

• mechanical Q benchmark (disorder + cross shield) on the Oh cavity; anisotropic elastic model test on the Oh mirror cell

• taper optimization (Oh: genetic algorithm on a g_0Q_o fitness) after the final shapes

• 2D phononic shield at the final breathing frequency (snowflake / cross); 2D snowflake sweep diamond_a_large paused at 6/45